Rice Grain Variety Classification

1. Project Overview

This project uses a Convolutional Neural Network (CNN) to classify rice grain images into five varieties: Arborio, Basmati, Ipsala, Jasmine, and Karacadag. The model learns visual features from images and predicts the corresponding rice variety automatically.

2. Objectives

- To classify five different varieties of rice grains using deep learning.
- To preprocess and augment images for model training.
- To evaluate the CNN model using accuracy, precision, recall, and F1-score.
- To analyze classification performance using a confusion matrix.

3. Dataset

The dataset is obtained from Kaggle.

Dataset link: https://www.kaggle.com/datasets/muratkokludataset/rice-image-dataset

The dataset contains five classes:

- Arborio
- Basmati
- Ipsala
- Jasmine
- Karacadag

A balanced subset of 500 images per class was selected, giving a total of 2,500 images.

4. Technologies Used

- Python
- TensorFlow / Keras
- NumPy
- Pandas
- Matplotlib
- Scikit-learn
- Google Kaggle Notebooks

5. Methodology

1. Load the rice grain image dataset.

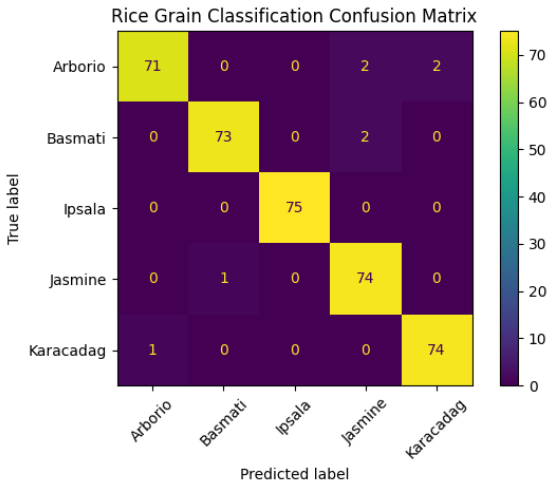
2. Select a balanced subset of images from each class.
3. Resize images to 128 × 128 pixels and normalize pixel values.
4. Split the dataset into 70% training, 15% validation, and 15% testing.
5. Apply data augmentation to training images.
6. Build and train a CNN model.
7. Evaluate the model using a classification report and confusion matrix.

6. Model Configuration

Parameter	Configuration
Model	Convolutional Neural Network (CNN)
Image size	128 × 128
Images per class	500

7. Results

Evaluation Set	Accuracy	Weighted F1-score	Interpretation
Training set	92.20%	0.9920	Shows how well the model learned from training images.
Validation split	98.40%	0.9840	Measures performance on unseen validation images during training.
Test set	98.00%	0.9800	Evaluates the model's performance on unseen rice grain images.





The model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. Training and validation accuracy/loss graphs are used to observe model performance.

8. Real-World Applications

- Rice grain variety identification
- Agricultural quality inspection
- Automated grain sorting
- Food processing and packaging

9. Limitations

The model is trained using a limited subset of images. Its performance may vary with different lighting conditions, backgrounds, and image quality. Further testing on real-world rice grain images is required.

10. Future Enhancements

- Train with a larger and more diverse dataset.
- Improve performance through hyperparameter tuning.
- Compare CNN with transfer learning models.
- Develop a simple web application for rice variety prediction.

11. Dataset Citation

Koklu, M. *Rice Image Dataset*. Kaggle.

https://www.kaggle.com/datasets/muratkokludataset/rice-image-dataset